

Cookie Crumbles: Breaking and Fixing Web Session Integrity

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Cookie Crumbles: Breaking and Fixing Web Session Integrity

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Cookies have a long history of vulnerabilities targeting their confidentiality and integrity. To address these issues, new mechanisms have been proposed and implemented in browsers and server-side applications. Notably, improvements to the Secure attribute and cookie prefixes aim to strengthen cookie integrity against network and same-site attackers, whereas SameSite cookies have been touted as the solution to CSRF. On the server, token-based protections are considered an effective defense for CSRF in the synchronizer token pattern variant. In this paper, we question the effectiveness of these protections and study the real-world security implications of cookie integrity issues, showing how security mechanisms previously considered robust can be bypassed, exposing Web applications to session integrity attacks such as session fixation and cross-origin request forgery (CORF). These flaws are not only implementation-specific bugs but are also caused by compositionality issues of security mechanisms or vulnerabilities in the standard. Our research contributed to 12 CVEs, 27 vulnerability disclosures, and updates to the cookie standard. It comprises (i) a thorough cross-browser evaluation of cookie integrity issues, that results in new attacks originating from implementation or specification inconsistencies, and (ii) a security analysis of the top 13 Web frameworks, exposing session integrity vulnerabilities in 9 of them. We discuss our responsible disclosure and propose practical mitigations.

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